

Meitu: Improving the selfie experience by beautifying people with AI technology

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Meitu uses Google Cloud to better understand its customers by collecting and analyzing data, so we can develop a user-friendly platform where people can digitally beautify themselves.

meitu美图

About Meitu

Meitu is a leading image processing and social networking platform in China driven by artificial intelligence. Its mission is to "let everyone become beautiful," with the help of easy-to-use virtual apps such as Meitu Xiuxiu and Meiyan camera, among others. As of December 2019, Meitu's image and community application matrix had been activated on more than 1.88 billion individual devices worldwide, with more than 282 million monthly active users in China and more than 724 million overseas users.

Google Cloud results

- Google Kubernetes Engine improves resource utilization with autoscaling capabilities to reduce 44% operational costs
- BigQuery analyzes user behavior to get more accurate user insights
- Firebase reduces product split testing time by 50% so that engineers can roll out

44% re
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million

Industries: Technology

Location: China

(<https://cloud.google.com/cloud-shell/>),
Google Cloud (<https://cloud.google.com/>)

(<https://marketingplatform.google.com/about/data-studio/>)
BigQuery (<https://cloud.google.com/bigquery/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Cloud Shell (<https://cloud.google.com/cloud-shell/>)

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Looker Studio
(<https://marketingplatform.google.com/about/data-studio/>)

Anthos (<https://cloud.google.com/anthos/>)

**new
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Social media and smartphones have made the act of taking a photo of yourself, or a “selfie,” part of daily life. Photo filters have become widely popular, allowing you to recreate an old Hollywood look in black and white, add sparkles to your photos, or even turn yourself into a cartoon puppy. In this era of “selfie culture,” Meitu (<https://meitu.com/en/>) was founded with a mission to help people digitally enhance their photos using technology.

Since its launch in 2008, Meitu has created a series of products centered around the theme of beauty, such as image editing software Meitu Xiuxiu, effects and filters app Meiyen Camera, short video platform Meipai, skin analyzer app Meitu Eve, and software development kit MeituGenius. With these innovative tools, the company is changing the way people create and share photos and has contributed to the growth of selfie culture in China. As of December 2019, the Meitu app had been downloaded on more than 1.88 billion independent devices worldwide, with more than 282 million monthly active users in China. And in the rest of the world, Meitu’s user base has grown to more than 724 million overseas users.

Contributing to this remarkable success is Meitu's Imaging & Vision Lab (MTlab), a research and development (R&D) department set up in 2010 and dedicated to technological innovation. Through its R&D in AI-related fields such as computer vision, deep learning, and computer graphics, MTlab has been actively developing its facial recognition technology, image segmentation, image enhancement, and image generation capabilities.

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—Eagle Lee, Meitu Eve
Senior Vice President,
Meitu

As Meitu’s overseas user base grew, the company’s technical team began to experience scaling issues, due to the limitations of its existing container service. “At that time, we felt we needed to simplify our IT infrastructure to meet our business needs, so we used a separate cloud provider as a cloud processing server for overseas clients,” explains Mingyang Song, head of Meitu’s Overseas Business Department. However, this arrangement meant that the team had to scale clusters manually, which was a complex and costly task.

Meitu Eve Senior Vice President Eagle Lee adds that when analyzing the image data, Meitu’s data analysts had to sort out the collected data, label it, and then use these labels for analysis. By late 2018, Meitu realized that it needed to simplify its backend processes, and for this, it turned to Google Cloud (<https://cloud.google.com/>).

“We want to bring the best user experience to our customers whenever they use our app,” says Lee. “To do this, we needed to reduce the technical layers required in our daily operations, save cost, and

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**Maximizing small machines and
small teams with Google
Kubernetes Engine**

One of the key benefits of moving to Google Cloud was its ability to use the autoscaling feature of Google Kubernetes Engine (/kubernetes-engine) (GKE).

"Google Kubernetes Engine helped us simplify the workflow and reduce operational costs by 44%, thanks to its auto-scaling feature that allows us to use small machines with a high bandwidth," says Lee. He adds that the load balancing feature of Google Cloud helped enhance Meitu's user experience because it uses the same infrastructure as Google Search and YouTube.

With autoscaling, Meitu has also been able to handle its day-to-day operations with its existing team of developers, and doesn't need to hire additional staff.

"The high degree of automation has enabled us to reduce our manpower by four people. Before adopting GKE, every application needed an engineer to maintain the operation. Now it only needs very few personnel to ensure that all applications run smoothly," says Lee.

Another benefit of migrating to Google Cloud is the increased rendering speed of its beauty camera. The auto-scaling function of GKE enables users to see their enhanced photos in just a few seconds, improving its overall user experience, which keeps them coming back. "The container technology and autoscaling technology of GKE is currently one of the best in the industry and has helped us save a lot on our operational expenses," says Song.

At the same time, Meitu uses Anthos (/anthos), a multicluster management service, to deploy and manage its containers around the world. Based on

the GitOps idea of Google Cloud, Anthos helps Meitu to swiftly deploy multiple Kubernetes clusters at the global level and release new applications to global users simultaneously. This not only reduces Meitu's maintenance workload but also decreases the error rate upon the release of new applications.



Understanding user behavior with BigQuery

Having first adopted BigQuery (/bigquery) in early 2018, Meitu has seen the benefits of the scalable cloud data warehouse. It uses BigQuery to analyze the number of clicks and what users do on its platform. With such a wide reach, Meitu users are not all in locations with good network conditions. This can affect the accuracy of the data collection process. However, the ability of BigQuery to gather data in real time and analyze it using its machine learning (ML) capabilities even from areas with poorer network conditions helps simplify the computing process for Meitu's data team.

"In the past, we had to build data query services manually, which was a lot of work, not to mention a little problematic when trying to reach users in poor network conditions. BigQuery solves this problem by doing it automatically."

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Turning creative ideas into creative apps with Firebase

As an app that runs on different mobile platforms, Meitu uses Firebase (<https://firebase.google.com/>) to ensure that its products work seamlessly across platforms. The versatility of Firebase also allows developers to input data from the app into BigQuery for user behavior analysis, so that it can better understand its customers' needs.

"With Firebase, the team is able to execute new ideas without worrying too much about the backend infrastructure. This means that our engineers can spend more time being creative rather than worrying about troubleshooting," says Song.

When rolling out new products, Meitu usually runs split testing by putting out two similar products to see which one works better in the intended market. Since the use of Firebase, it has reduced the split testing time by 50%. "Previously, our split testing took one and a half to two months before we could obtain representative amounts of data to make sense of it. This has been reduced to about four weeks since the use of Firebase," Song adds.

With Firebase SDK, engineers can collect the application usage data without writing any additional codes. According to Lee, the Firebase SDK is also very user friendly. It automatically records when users have basic interactions with the application, allowing developers to easily find information such as the number of times users open

the app and how many active users there are in a specific period of time.

Meitu also uses Cloud Shell (/shell) to manage its Google Cloud resources. As Lee says, "We find this tool convenient and easy to use as we can access it directly on the browser. On top of that, the built-in authorization for access to projects and resources gives employees full access to their relevant projects, removing the need for an intermediary server, saving cost."

What's next: enhancing selfies even further with AL and ML

As the world moves towards the 5G era, Meitu plans to introduce a series of new products compatible with 5G networks to meet the market demands. Lee says, "For 5G applications, this is an upgrade of the internet service infrastructure. It can enhance the performance in some application scenarios of Meitu services, such as the advanced beautification feature that relies on augmented reality applications. MTlab has also been developing new projects related to artificial intelligence (AI) and ML. For example, the Meitu Xiuxiu painting robot launched a year ago is now one of the world's first products to use AI in painting."

"As a developer, I think the simplicity of Google Cloud products and its ability to be easily applied to an existing workflow is invaluable," says Lee. "From the deployment stage to using tools such as Firebase and BigQuery to optimize workflows, this whole experience has been pleasant." He adds that

before migrating to Google Cloud, engineers had to write scripts, interfaces, and make sure they all corresponded to the right applications manually. With Google Cloud, all these tools are integrated automatically, allowing Meitu to eliminate all of these steps and improve business efficiency.

In view of the close relationship between cloud beautification and cloud photo processing, Meitu will continue to leverage the capabilities of augmented reality and ML to develop innovative new products. "We hope to continue fulfilling our mission of helping people enjoy beauty and creativity," says Lee, "and of leading new aesthetic trends."

